

THE ZERO PARADOX

ZP-M: Kleene-Ordinal Bridge

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Synthesis bridge. All theorems §I–§IV proved sorry-free in Lean 4. Closes the hfp free hypothesis (ZP-L) and the snapEmbed type bridge. Axiom footprint: [propext, Classical.choice, Quot.sound] throughout.

ZP-M bridges the two prior layers — ZP-K (computational grounding via Kleene's second recursion theorem) and ZP-L (ordinal snap threshold at ϵ_0) — with two formal results. First, the free hypothesis hfp in snap_exactly_at_epsilon_zero (ZP-L §VI) is proved to follow from a minimal alignment hypothesis plus monotonicity: hfp is not an independent condition. Second, the canonical type bridge snapEmbed : MachinePhase $\rightarrow$ $\mathbb{Z}_2$ formalizes the structural triangle connecting the pre-snap state c_0 (read as the Kleene quine, a KleeneStructure requirement in ZP-K, not a theorem), the ordinal snap at ϵ_0 (ZP-L), and the 2-adic limit 0 (ZP-B).

The closing result (§IV) records that both the Kleene recursion theorem and the ordinal fixed-point $\epsilon_0 = \omega^{\epsilon_0}$ are instances of the same diagonalization pattern: a self-referential operation produces a forced fixed point. The two domains — computability theory (codes and Gödel numbers) and ordinal theory (ω -tower iteration) — use no shared mathematical machinery, but the structural schema is identical. Remark R-M.1 tracks what this means for DA-1 Path 2 and the boundary of the diagonalization frame.

Section I: The snapEmbed Morphism

snapEmbed is the canonical type bridge from ZP-E's machine-phase state space to ZP-B's 2-adic integers. The pre-snap state c_0 maps to 1 (a 2-adic unit); the snap state c_1 maps to 0 (the 2-adic limit of the tower encodings). Under multiplication on $\mathbb{Z}_2$, 0 is absorbing — mirroring exactly the join structure on MachinePhase, where c_1 is the absorbing element.

Within the ZP framework, $0 \in \mathbb{Z}_2$ plays the role of $\perp$ — it is the 2-adic valuation limit and the fixed point of multiplication by 2. snapEmbed makes this modelling commitment concrete: c_1 maps to 0 because c_1 IS the object at infinite 2-adic depth. This is a modelling commitment, not a ring-theoretic theorem.

Definition: snapEmbed (Incompleteness.lean §I)

snapEmbed : MachinePhase $\rightarrow$ $\mathbb{Z}_2$

snapEmbed $c_0 = 1$ (pre-snap: 2-adic unit)

snapEmbed $c_1 = 0$ (snap state: 2-adic zero = 2-adic limit of tower)

The morphism property: join on MachinePhase (c_1 is absorbing under join) corresponds to multiplication on $\mathbb{Z}_2$ (0 is absorbing under $\times$). Both structures share the absorbing-element pattern — the bridge is structural, not incidental.

Theorem: snapEmbed_injective (Incompleteness.lean §I)

Function.Injective snapEmbed

c_0 and c_1 map to distinct 2-adic integers ($1 \neq 0$ in $\mathbb{Z}_2$).

Proof: case split on MachinePhase; simp_all closes each case.

Lean purity: [propext, Classical.choice, Quot.sound]. ✓

Theorem: snapEmbed_mul_morphism (Incompleteness.lean §I)

$\forall a\ b : \text{MachinePhase}, \text{snapEmbed} (\text{join } a\ b) = \text{snapEmbed } a \times \text{snapEmbed } b$

snapEmbed sends the join operation to multiplication.

Under this map, the absorbing-element structure is preserved:

$\text{join } c_1\ x = c_1$ for all $x \leftrightarrow 0 \times y = 0$ for all $y \in \mathbb{Z}_2$.

Proof: case split; simp [snapEmbed].

Lean purity: [propext, Classical.choice, Quot.sound]. ✓

Lemma: snapEmbed_c0_val (Incompleteness.lean §I)

$(\text{snapEmbed } c_0).\text{valuation} = 0$

c_0 maps to a 2-adic unit: valuation 0, norm 1.

Proof: simp via PadicInt.valuation_one.

Lean purity: [propext, Classical.choice, Quot.sound]. ✓

Lemma: snapEmbed_c1_dvd (Incompleteness.lean §I)

$\forall n : \mathbb{N}, (2 : \mathbb{Z}_2)^n \mid \text{snapEmbed } c_1$

c_1 maps to 0, which is divisible by all powers of 2.

$0 \in \mathbb{Z}_2$ is divisible by 2^n for every n — infinite 2-adic depth.

This is the formal signature of $\perp = \{\perp\}$: infinite 2-adic valuation.

Proof: simp [snapEmbed_c1].

Lean purity: [propext, Classical.choice, Quot.sound]. ✓

Remark: snapEmbed vs the Identification Conjecture

snapEmbed establishes the morphism property ($\text{join} \mapsto \times$) and the absorbing-element correspondence. It does not derive the identification $\varepsilon_0 \leftrightarrow \perp$ as a type-theoretic theorem. The canonical threshold map $\text{Ordinal} \rightarrow \text{MachinePhase}$ is order-non-decreasing in ZP-A's join order (snap_map_mono), and no map $\text{Ordinal} \rightarrow \text{MachinePhase}$ is compatible with the $\text{CNF} \rightarrow \mathbb{Z}_2$ map along the tower: through snapEmbed the square fails at every stage $n \geq 1$ (the examples after c1_epsilon_zero_identification in ZeroParadox/Ordinal/Gentzen.lean, and in ZeroParadox/Ordinal/Incompleteness.lean § II).

What the bridge achieves: c_1 (the snap state assigned at ε_0) maps to 0 (the 2-adic limit) under snapEmbed. The triangle is formally exhibited. The conjectured stronger bridge — Classical.choice forced by ZP metric collapse — is deferred to §V (future work).

Section II: Closing the hfp Gap

snap_exactly_at_epsilon_zero (ZP-L §VI) carries a free hypothesis hfp asserting that for any map $\phi : \text{Ordinal} \rightarrow \text{MachinePhase}$, every fixed point of $\omega^\cdot$ maps to c_1 . This hypothesis is required for the upper-bound direction of the minimality argument (ε_0 is the first snap point). The canonical witness in epsilon_zero_snap_canonical (ZP-L §VII) satisfies hfp directly by case analysis, but hfp is not derived from the structural axioms.

ZP-M §II closes this gap. Given monotonicity and $\phi \varepsilon_0 = c_1$ as a minimal alignment hypothesis, hfp follows. The argument: for any fixed point α of $\omega^\cdot$, epsilonZero_le_fixedPoint gives $\varepsilon_0 \leq \alpha$; monotonicity then chains $\phi \varepsilon_0 \leq \phi \alpha$; substituting c_1 and using c_1 's absorbing property gives $\phi \alpha = c_1$.

Theorem: hfp_from_epsilon_zero (Incompleteness.lean §II)

For any $\phi : \text{Ordinal} \rightarrow \text{MachinePhase}$ satisfying:

hmono: $\forall \alpha \leq \beta, \text{join} (\phi \alpha) (\phi \beta) = \phi \beta$

h ε_0 : $\phi \text{ epsilonZero} = c_1$

we have: $\forall \alpha, \omega^\cdot \alpha = \alpha \rightarrow \phi \alpha = c_1$.

Proof: epsilonZero_le_fixedPoint gives $\varepsilon_0 \leq \alpha$; hmono gives $\text{join } c_1 (\phi \alpha) = \phi \alpha$; $\text{join } c_1 x = c_1$ for all x (absorbing property of c_1), so $c_1 = \phi \alpha$.

Lean purity: [propext, Classical.choice, Quot.sound]. ✓

Theorem: snap_unconditional (Incompleteness.lean §II)

For any ϕ satisfying hmono, h0 (tower stages $\mapsto c_0$), and h ε_0 ($\phi \varepsilon_0 = c_1$):

$\phi \varepsilon_0 = c_1 \wedge \forall \alpha, \phi \alpha = c_1 \rightarrow \varepsilon_0 \leq \alpha$.

ε_0 is the minimal snap threshold under minimal hypotheses.

Theorem: snap_unconditional (Incompleteness.lean §II)

Proof: snap_exactly_at_epsilon_zero with hfp supplied by hfp_from_epsilon_zero.

Lean purity: [propext, Classical.choice, Quot.sound]. ✓

What §II achieves: hfp is no longer a free hypothesis in the minimality theorem. Given $\phi \varepsilon_0 = c_1$ (the alignment condition: the map snaps at ε_0) plus monotonicity, the full minimality result follows from the ordinal structure alone. hfp is a derived consequence, not an additional commitment.

Section III: The Kleene-Ordinal Triangle

ZP-K established: $c_0 = \perp$ is a Quine atom ($\text{da1_closed_concrete} : \text{IsQuineAtom } (\perp : \text{MachinePhase})$). ZP-L established: the canonical snap map assigns c_1 exactly at ε_0 , with tower encodings in $\mathbb{Z}_2$ converging to 0. The triangle connects all three objects formally:

Edge	Objects connected	Formal content
$A \leftrightarrow C$ (ordinal snap)	$\varepsilon_0 \leftrightarrow c_1$	Canonical snap map: stages below $\varepsilon_0 \mapsto c_0$; $\varepsilon_0 \mapsto c_1$ (ZP-L §VII)
$A \leftrightarrow B$ (2-adic convergence)	$\text{Tower} \rightarrow 0 \in \mathbb{Z}_2$	$\text{cnfToZp2}(\text{towerNONote } n) \rightarrow 0 \text{ in } \mathbb{Z}_2$ (ZP-L §IV)
$B \leftrightarrow C$ (type bridge)	$c_1 \leftrightarrow 0 \in \mathbb{Z}_2$	$\text{snapEmbed } c_1 = 0$ (§I above)

Theorem: snap_state_zp2_is_zero (Incompleteness.lean §III)

$\text{snapEmbed } ((\lambda \alpha : \text{Ordinal}, \text{if } \alpha < \varepsilon_0 \text{ then } c_0 \text{ else } c_1) \varepsilon_0) = 0$

The canonical snap map assigns c_1 at ε_0 ; snapEmbed sends c_1 to 0.

The snap state at the ordinal threshold maps to the 2-adic zero.

Proof: simp (the if-condition evaluates to False at $\alpha = \varepsilon_0$).

Lean purity: [propext, Classical.choice, Quot.sound]. ✓

Theorem: zpm_triangle (Incompleteness.lean §III)

All three edges of the triangle, co-proved:

$(A \leftrightarrow C) \forall n : \mathbb{N}, \text{fundamentalSeq } n < \varepsilon_0$

$(A \leftrightarrow C) (\lambda \alpha, \text{if } \alpha < \varepsilon_0 \text{ then } c_0 \text{ else } c_1) \varepsilon_0 = c_1$

$(A \leftrightarrow B) \text{Filter.Tendsto } (\lambda n, \text{cnfToZp2 } (\text{towerNONote } n)) \text{Filter.atTop } (\text{nhds } 0)$

$(B \leftrightarrow C) \text{snapEmbed } (\lambda \alpha, \text{if } \alpha < \varepsilon_0 \text{ then } c_0 \text{ else } c_1) \varepsilon_0 = 0$

Theorem: zpm_triangle (Incompleteness.lean §III)

Proof: $\langle \text{epsilonZero_tower_lt, if_neg (lt_irrefl } \epsilon_0), \text{tower_converges_to_zero, snap_state_zp2_is_zero} \rangle$.

Lean purity: [propxt, Classical.choice, Quot.sound]. ✓

Section IV: Shared Diagonalization Pattern

Both the Kleene recursion theorem (ZP-K) and the ordinal fixed-point ϵ_0 (ZP-L) arise from the same structural schema: a self-referential operation has a forced fixed point. In computability theory, the diagonal applies to codes and their Gödel numbers. In ordinal theory, the diagonal applies to the tower iteration $\alpha \mapsto \omega^\alpha$. The two domains share no mathematical machinery, but the diagonalization pattern is identical.

Domain	Self-referential operation	Forced fixed point
Computability (ZP-K)	$\text{eval } c = \text{selfApply } c$ (c runs on its own Gödel number)	$\exists c : \text{Code}, \text{IsComputationalQuine } c$
Ordinal theory (ZP-L)	$\omega^\alpha = \alpha$ (α is a fixed point of ω -exponentiation)	$\epsilon_0 = \text{nfp } (\omega^\cdot) 0$, the least such α

Theorem: both_fixed_points_exist (Incompleteness.lean §IV)

$(\exists c : \text{Code}, \forall n, \text{eval } c \ n = \text{eval } c \ (\text{encode } c + n)) \wedge$

$(\exists \alpha : \text{Ordinal}, \omega^\alpha = \alpha \wedge \forall \beta, \omega^\beta = \beta \rightarrow \alpha \leq \beta)$

Both diagonalization patterns produce fixed points, co-proved in one formal context.

Left: Kleene quine (period = Gödel number of c), via `computational_quine_exists`.

Right: $\epsilon_0 = \omega^{\epsilon_0}$, minimal; via `epsilonZero_fixedPoint` and `epsilonZero_le_fixedPoint`.

The co-existence is the formal content: two forced fixed points, two domains, one theorem.

Lean purity: [propxt, Classical.choice, Quot.sound]. ✓

Remark R-M.1: DA-1 Path 2 and the Limits of the Diagonalization Frame

`both_fixed_points_exist` establishes that Kleene diagonalization (a code on its own Gödel number) and ordinal diagonalization ($\epsilon_0 = \omega^{\epsilon_0}$, least fixed point) are the same structural pattern in two domains. Both are textbook instances of the diagonalization schema.

L-INF (ZP-C §II: $\perp$ has divergent surprisal — unbounded information content, incompressible by any finite description) is structurally analogous. Whether it fits the same diagonalization schema formally is the open question.

The formally unconnected instance is L-INF. This gap is DA-1 Path 2 — the informational bridge. The reason Path 2 resisted formalization is now visible from this layer: L-INF is a measure-theoretic statement (Shannon entropy, probability distributions) while the Kleene quine is a computability-theoretic statement (partial recursive functions, Gödel encoding). The two frameworks share no mathematical machinery.

Remark R-M.1: DA-1 Path 2 and the Limits of the Diagonalization Frame

The diagonalization frame established here provides the common conceptual ground, but the formal bridge requires the Kolmogorov complexity connection: incompressibility is the concept that lives in both worlds simultaneously. That bridge is outside the current Lean scope, pending AIT infrastructure not yet in Mathlib.

DA-1 Path 2 was recharacterized in ZP-E/ZP-K as a foundational commitment rather than a missing proof. The diagonalization frame here makes the boundary precise: the gap is not a skipped proof step but a genuine framework separation — measure-theoretic surprisal on one side, computability-theoretic encoding on the other, with no shared machinery in current Mathlib.

Theorem Summary

Theorem	Section	Status
snapEmbed_injective	§I	Proved ✓
snapEmbed_mul_morphism	§I	Proved ✓
snapEmbed_c0_val	§I	Proved ✓
snapEmbed_c1_dvd	§I	Proved ✓
hfp_from_epsilon_zero	§II	Proved ✓
snap_unconditional	§II	Proved ✓
snap_state_zp2_is_zero	§III	Proved ✓
zpm_triangle	§III	Proved ✓
both_fixed_points_exist	§IV	Proved ✓

Axiom Purity

All theorems carry axiom footprint: [propext, Classical.choice, Quot.sound].

These are standard Mathlib infrastructure axioms, inherited from ordinal theory (ZP-L), 2-adic analysis (ZP-B), and computability theory (ZP-K).

Classical.choice is load-bearing in the ordinal fixed-point (nfp) and present in the computability fixed-point (Kleene's theorem). In the computability layer Classical.choice is reached through Mathlib's numbering of program codes (Denumerable Code): it is present even in a computable instance with a constant code. ZP-K's machinePhaseKleene also picks botCode with Classical.choose, which is what makes that instance noncomputable. Whether the numbering's footprint is essential is not measured. ZP-K Section IV tabulates the measured footprints.

Zero sorry in Incompleteness.lean. Verified: lake build, May 2026.

End of ZP-M | Kleene–Ordinal Bridge | snapEmbed type bridge | hfp gap closed | zpm_triangle co-proved | both_fixed_points_exist | R-M.1: DA-1 Path 2 boundary | All Incompleteness.lean theorems verified. Axioms: [propext, Classical.choice, Quot.sound].